



```

name: InactivityAndMarriage
log: C:\Users\hansg\Documents\working\SHARE data 5\output.smcl
log type: smcl
opened on: 30 Jun 2026, 11:06:03

```

```

1 .
2 . use "C:\Users\hansg\Documents\working\SHARE data 5\dataset.dta"

3 .
4 . asdoc, text(\par \qc The Influence of Health, Marriage and Cohabitation on Physical
> Inactivity: Evidence from SHARE 2015 to 2022 for 12 European Countries) fs(12) save
> (report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

5 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

6 .
7 . ***variables preparation and data description
8 .
9 . asdoc, text(\page) save(report.doc) append
Click to Open File: report.doc

10.
11. *panel level id
12. egen id= group(mergeid)

13. drop *_f

14. drop if phinact== -99
(24,644 observations deleted)

15.
16. xtset id Qyear

Panel variable: id (unbalanced)
Time variable: Qyear, 2015 to 2022, but with gaps
Delta: 1 unit

17.
18. asdoc xttrans phinact, freq save(report.doc) append

```

Physical inactivity	Physical inactivity		Total
	0	1	
0	51,749 91.29	4,940 8.71	56,689 100.00
1	2,527 43.80	3,243 56.20	5,770 100.00
Total	54,276 86.90	8,183 13.10	62,459 100.00

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19.
```

20. tabulate country phinact

Country identifier	Physical inactivity		Total
	No	Yes	
Austria	6,082	736	6,818
Germany	9,414	838	10,252
Sweden	8,500	476	8,976
Spain	7,904	1,921	9,825
Italy	8,582	3,068	11,650
France	7,733	1,034	8,767
Denmark	8,288	515	8,803
Greece	10,995	1,215	12,210
Switzerland	6,360	327	6,687
Belgium	10,625	1,569	12,194
Czechia	9,139	1,162	10,301
Poland	3,789	862	4,651
Total	97,411	13,723	111,134

21.

22. generate mst=0

23. replace mst=1 if mstat==3 | mstat==5
(11,008 real changes made)

24. replace mst=2 if mstat==4
(6,113 real changes made)

25. replace mst=3 if mstat==6
(17,718 real changes made)

26. label define msL 0 "Officially with partner" 1 "Away from partner" 2 "Never married"
> 3 "Lost partner"

27. label values mst msL

28.

29. spearman mst single

Number of observations = **111,134**
Spearman's rho = **0.8537**

Test of H0: **mst** and **single** are independent
Prob = **0.0000**

30.

31. generate change=0

32. replace change=1 if mst==0
(76,295 real changes made)

33. bysort id: generate changed = change - change[_n-1]
(48,675 missing values generated)

34. replace changed=changed+1
(62,459 real changes made)

35. replace changed=1 if changed==.
(48,675 real changes made)

36. label define chL 0 "No longer officially together" 1 "All other situations" 2 "Newly
> officially together"

37. label values changed chL

38. asdoc tabulate changed, save(report.doc) append

changed	Freq.	Percent	Cum.
No longer officially together	2,072	1.86	1.86
All other situations	108,733	97.84	99.70
Newly officially together	329	0.30	100.00
Total	111,134	100.00	

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39.

40. bysort id: generate schanged=single - single[_n-1]
(48,675 missing values generated)

41. replace schanged=0 if schange==.
(48,675 real changes made)

42. replace schanged=schanced+1
(111,134 real changes made)

43. label define schL 0 "Hooked up" 1 "Unchanged" 2 "Back single"

44. label values schanged schL

45. asdoc tabulate schanged, save(report.doc) append

schanged	Freq.	Percent	Cum.
Hooked up	430	0.39	0.39
Unchanged	107,804	97.00	97.39
Back single	2,900	2.61	100.00
Total	111,134	100.00	

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46.

47. generate mortg=0

48. replace mortg=1 if mort>0
(17,873 real changes made)

49. label define mlab 0 "No mortgage" 1 "Mortgage"

50. label values mortg mlab

51.

52. codebook phinact sphus thinc mst changed schanged cjs age yedu gender country bmi eu
> rod, compact

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	111134	2	.1234816	0	1	Physical inactivity
sphus	111134	5	3.114771	1	5	Self-perceived health - US scale
thinc	111134	56252	32869.01	0	7399056	Total household income - Version
> A						
mst	111134	4	.6873504	0	3	
changed	111134	3	.9843162	0	2	
schanged	111134	3	1.022225	0	2	
cjs	111134	7	2.24762	-99	97	Current job situation
age	111134	47	70.20186	50	96	Age
yedu	111134	30	11.01304	0	30	Years of education
gender	111134	2	1.561619	1	2	Gender
country	111134	12	18.22554	11	29	Country identifier
bmi	111134	3652	26.82758	12.484	98.61933	Body mass index

eurod	111134	13	2.380928	0	12	EURO depression scale
-------	--------	----	----------	---	----	-----------------------

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53.
54. drop if cjs==99
    (1,089 observations deleted)
55. drop if yedu==9997
    (0 observations deleted)
56.
57. winsor2 thinc, replace cuts(0 99)
58.
59. codebook phinact sphus thinc mst changed schanged cjs age yedu gender country bmi eu
    > rod, compact

```

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	110045	2	.119233	0	1	Physical inactivity
sphus	110045	5	3.106974	1	5	Self-perceived health - US sca
> le thinc	110045	54718	31725.99	0	193799.5	Total household income - Versi
> on A mst	110045	4	.6757599	0	3	
changed	110045	3	.9847608	0	2	
schanged	110045	3	1.021437	0	2	
cjs	110045	6	3.249562	1	97	Current job situation
age	110045	47	70.09174	50	96	Age
yedu	110045	30	11.02087	0	30	Years of education
gender	110045	2	1.560752	1	2	Gender
country	110045	12	18.21905	11	29	Country identifier
bmi	110045	3634	26.8345	12.86854	98.61933	Body mass index
eurod	110045	13	2.371121	0	12	EURO depression scale

```

60.
61. global names thinc
62. foreach num of numlist 1(1)1 {
    2.     local name: word `num' of $names
    3.     bysort country: egen mean`name' = mean(`name') if `name'!=0
    4.     generate T=1
    5.     replace T=0 if `name'==0
    6.     replace T=2 if `name'>mean`name'
    7.     drop mean`name'
    8.     drop `name'
    9.     rename T `name'
    10. }
    (1,712 missing values generated)
    (1,712 real changes made)
    (41,972 real changes made)

```

```

63.

```

```

64. label define thL 0 "No income" 1 "Below country average income" 2 "Above country ave
> rage income"

65. label values thinc thL

66.

67. keep gali sphus thinc mst mstat changed schanged cjs age yedu gender country single
> id Qyear my_wgt phinact bmi eurod

68.

69. *asdoc spearman phinact gali sphus thinc single mst changed schanged cjs age yedu ge
> nder country bmi eurod, save(report.doc) append

70.

71. label copy cjs cjsLab

72. label list cjsLab
cjsLab:
    -99 Not applicable
      1 Retired
      2 Employed or self-employed
      3 Unemployed
      4 Permanently sick
      5 Homemaker
     97 Other

73. replace cjs=6 if cjs==97
    (1,823 real changes made)

74. label define cjsLab 6 "Other", add

75. label values cjs cjsLab

76. tabulate cjs, nolab

```

Current job situation	Freq.	Percent	Cum.
1	73,119	66.44	66.44
2	20,471	18.60	85.05
3	1,877	1.71	86.75
4	2,700	2.45	89.21
5	10,055	9.14	98.34
6	1,823	1.66	100.00
Total	110,045	100.00	

```

77.

78. bysort id: egen Mundlak_OthC_age=mean(age)

79. bysort id: egen Mundlak_OthC_yedu=mean(yedu)

80. bysort id: egen Mundlak_OthC_bmi=mean(bmi)

81. bysort id: egen Mundlak_OthC_eurod=mean(eurod)

82.

83.

84.

85. *program for calculating Mundlak_Means
86. program define Mundlak_Means
    1.      *syntax varname

```

```

87.      global variab changed schanged mst single sphus gali cjs thinc
      2.      foreach V of numlist 1(1)8 {
      3.          local va: word `V' of $variab
      4.          tab `va', gen(_dum_)
      5.          levelsof `va', _local(levels)
      6.          local i = 1
      7.          foreach l of local levels {
      8.              local lbl : label (`va') `l'
      9.              local clean = ustrword("`lbl'",1)
     10.              rename _dum_`i' `va'_clean'
     11.              local ++i
     12.          }
     13.          foreach v of varlist `va'_* {
     14.              if `V'==1 {
     15.                  bysort id: egen Mundlak_C_`v' = mean(`v')
>
     16.              }
     17.              if `V'==2 {
     18.                  bysort id: egen Mundlak_sC_`v' = mean(`v')
>
     19.              }
     20.              if `V'==3 {
     21.                  bysort id: egen Mundlak_Mar_`v' = mean(`v')
>
     22.              }
     23.              if `V'==4 {
     24.                  bysort id: egen Mundlak_Sin_`v' = mean(`v')
>
     25.              }
     26.              if `V'==5 {
     27.                  bysort id: egen Mundlak_Sph_`v' = mean(`v')
>
     28.              }
     29.              if `V'==6 {
     30.                  bysort id: egen Mundlak_Gal_`v' = mean(`v')
>
     31.              }
     32.              if `V'==7 | `V'==8 {
     33.                  bysort id: egen Mundlak_Oth_`v' = mean(`v')
     34.              }
     35.          }
     36.      }
     37. end

```

88.

89. Mundlak_Means

changed	Freq.	Percent	Cum.
No longer officially together	2,005	1.82	1.82
All other situations	107,712	97.88	99.70
Newly officially together	328	0.30	100.00
Total	110,045	100.00	

0 1 2

schanged	Freq.	Percent	Cum.
Hooked up	429	0.39	0.39
Unchanged	106,828	97.08	97.47
Back single	2,788	2.53	100.00
Total	110,045	100.00	

0 1 2

mst	Freq.	Percent	Cum.
Officially with partner	76,018	69.08	69.08
Away from partner	10,875	9.88	78.96
Never married	5,967	5.42	84.38
Lost partner	17,185	15.62	100.00
Total	110,045	100.00	

0 1 2 3

Single	Freq.	Percent	Cum.
No	77,574	70.49	70.49
Yes	32,471	29.51	100.00
Total	110,045	100.00	

0 1

Self-perceived health - US scale	Freq.	Percent	Cum.
Excellent	7,401	6.73	6.73
Very good	21,370	19.42	26.14
Good	42,951	39.03	65.18
Fair	28,702	26.08	91.26
Poor	9,621	8.74	100.00
Total	110,045	100.00	

1 2 3 4 5

Limitation with activities	Freq.	Percent	Cum.
Not limited	60,290	54.79	54.79
Limited	49,755	45.21	100.00
Total	110,045	100.00	

0 1

Current job situation	Freq.	Percent	Cum.
Retired	73,119	66.44	66.44
Employed or self-employed	20,471	18.60	85.05
Unemployed	1,877	1.71	86.75
Permanently sick	2,700	2.45	89.21
Homemaker	10,055	9.14	98.34
Other	1,823	1.66	100.00
Total	110,045	100.00	

1 2 3 4 5 6

thinc	Freq.	Percent	Cum.
No income	1,712	1.56	1.56
Below country average income	66,361	60.30	61.86
Above country average income	41,972	38.14	100.00
Total	110,045	100.00	

0 1 2

```

90.
91. generate iG1=1.changed_No#eurod
92. generate iG2=1.changed_All#eurod
93. generate iG3=1.changed_Newly#eurod
94.
95. generate isG1=1.schanged_Hooked#eurod
96. generate isG2=1.schanged_Unchanged#eurod
97. generate isG3=1.schanged_Back#eurod
98.
99. foreach n of numlist 1(1)3 {
    2.         bysort id: egen Mundlak_Gi_`n'=mean(iG`n')
    3. }

100
101 foreach n of numlist 1(1)3 {
    2.         bysort id: egen Mundlak_sGi_`n'=mean(isG`n')
    3. }

102
103
104 ***descriptives
105
106 asdoc codebook phinact sphus thinc mst changed schanged cjs age yedu gender country
    > bmi eurod, compact save(report.doc) append

```

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	110045	2	.119233	0	1	Physical inactivity
sphus	110045	5	3.106974	1	5	Self-perceived health - US sca
> le						
thinc	110045	3	1.36585	0	2	
mst	110045	4	.6757599	0	3	
changed	110045	3	.9847608	0	2	
schanged	110045	3	1.021437	0	2	
cjs	110045	6	1.74206	1	6	Current job situation
age	110045	47	70.09174	50	96	Age
yedu	110045	30	11.02087	0	30	Years of education
gender	110045	2	1.560752	1	2	Gender
country	110045	12	18.21905	11	29	Country identifier
bmi	110045	3634	26.8345	12.86854	98.61933	Body mass index
eurod	110045	13	2.371121	0	12	EURO depression scale

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```

107
108 graph bar (count), over(mst, label(angle(90) labsize(small))) over(Qyear, label(angl
    > e(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(small))
    > ylabel(0(2500)40000,angle(0) labsize(small) grid)

109 graph export descrip1.png, replace height(2400)
    file descrip1.png saved as PNG format

```



```

110 graph bar (count), over(single, label(angle(90) labsize(small))) over(Qyear, label(a
> ngle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(sma
> l)) ylabel(0(2500)42500,angle(0) labsize(small) grid)

111 graph export descrip2.png, replace height(2400)
file descrip2.png saved as PNG format

112 graph bar (count), over(phinact, label(angle(90) labsize(small))) over(Qyear, label(
> angle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(sma
> ll)) ylabel(0(2500)52500,angle(0) labsize(small) grid)

113 graph export descrip3.png, replace height(2400)
file descrip3.png saved as PNG format

114 graph bar (count), over(sphus, label(angle(90) labsize(small))) over(Qyear, label(an
> gle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(smal
> )) ylabel(0(2500)20000,angle(0) labsize(small) grid)

115 graph export descrip4.png, replace height(2400)
file descrip4.png saved as PNG format

116
117 asdoc tabulate phinact, save(report.doc) append

```

Physical inactivity	Freq.	Percent	Cum.
No	96,924	88.08	88.08
Yes	13,121	11.92	100.00
Total	110,045	100.00	

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```

118
119 ***regressions
120
121 xtset id Qyear, yearly

Panel variable: id (unbalanced)
Time variable: Qyear, 2015 to 2022, but with gaps
Delta: 1 year

122
123 xtlogit phinact age yedu i.gender i.thinc i.mst ib1.changed i.gali i.cjs c.bmi c.eur
> od i.country

Fitting comparison model:

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33239.196
Iteration 2: Log likelihood = -31474.668
Iteration 3: Log likelihood = -31437.723
Iteration 4: Log likelihood = -31437.592
Iteration 5: Log likelihood = -31437.592

Fitting full model:

tau = 0.0 Log likelihood = -31437.592
tau = 0.1 Log likelihood = -31263.473
tau = 0.2 Log likelihood = -31098.19
tau = 0.3 Log likelihood = -30953.791
tau = 0.4 Log likelihood = -30850.606
tau = 0.5 Log likelihood = -30824.183
tau = 0.6 Log likelihood = -30940.648

Iteration 0: Log likelihood = -30824.183
Iteration 1: Log likelihood = -30271.77
Iteration 2: Log likelihood = -30234.943
Iteration 3: Log likelihood = -30234.55
Iteration 4: Log likelihood = -30234.55

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-30234.55**

Wald chi2(29) = **7017.88**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0955753	.0023314	40.99	0.000	.0910058
> .1001449	yedu	-.0288965	.0040766	-7.09	0.000	-.0368865
> -.0209065	gender					
	Female	.0184915	.035508	0.52	0.603	-.0511028
> .0880859	thinc					
	Below country average income	-.4534341	.0961682	-4.72	0.000	-.6419204
> -.2649478	Above country average income	-.6560687	.0990685	-6.62	0.000	-.8502394
> -.4618979	mst					
	Away from partner	.1015575	.0585961	1.73	0.083	-.0132888
> .2164038	Never married	.3300291	.0717943	4.60	0.000	.1893149
> .4707433	Lost partner	.090229	.0451345	2.00	0.046	.0017671
> .178691	changed					
	No longer officially together	-.2940148	.0923621	-3.18	0.001	-.4750411
> -.1129885	Newly officially together	.3953265	.2439247	1.62	0.105	-.0827571
> .8734101	gali					
	Limited	1.417282	.033636	42.14	0.000	1.351357
> 1.483208	cjs					
	Employed or self-employed	.2222638	.0602958	3.69	0.000	.1040861
> .3404414	Unemployed	.5877893	.1180766	4.98	0.000	.3563634
> .8192153	Permanently sick	1.690957	.0762829	22.17	0.000	1.541445
> 1.840469	Homemaker	.1891099	.0535201	3.53	0.000	.0842125
> .2940072	Other	.6676439	.0942118	7.09	0.000	.4829922
> .8522956	bmi	.0477364	.0031852	14.99	0.000	.0414936
> .0539792	eurod	.192974	.0061202	31.53	0.000	.1809786
> .2049693	country					
	Germany	-.2318761	.0882329	-2.63	0.009	-.4048095
> -.0589427						

> -.7665399	Sweden	-.9586265	.0980052	-9.78	0.000	-1.150713
> 1.042715	Spain	.8843885	.0807802	10.95	0.000	.7260621
> 1.989613	Italy	1.835259	.0787537	23.30	0.000	1.680904
> .5087977	France	.3393567	.0864511	3.93	0.000	.1699158
> -.2419429	Denmark	-.435573	.0987927	-4.41	0.000	-.6292031
> .3945341	Greece	.2305772	.083653	2.76	0.006	.0666203
> -.589939	Switzerland	-.8006815	.1075237	-7.45	0.000	-1.011424
> .5678298	Belgium	.4085186	.0812827	5.03	0.000	.2492075
> .4321668	Czechia	.2685154	.0834972	3.22	0.001	.1048639
> 1.017698	Poland	.8288242	.0963659	8.60	0.000	.6399504
> -11.67617	_cons	-12.17746	.2557661	-47.61	0.000	-12.67876
> 1.095025	/lnsig2u	1.021505	.0375108			.9479856
> 1.728947	sigma_u	1.666545	.0312567			1.606395
> .4760627	rho	.4577654	.0093108			.4395812

LR test of rho=0: chibar2(01) = 2406.08

Prob >= chibar2 = 0.000

124 estimates store RE

125 xtlogit phinact age yedu i.gender i.thinc i.mst ibl.changed i.gali i.cjs c.bmi c.eur

> od i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or all negative outcomes.

note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5823.4647

Iteration 1: Log likelihood = -5415.1378

Iteration 2: Log likelihood = -5413.4208

Iteration 3: Log likelihood = -5413.4207

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = 17,262

Number of groups = 5,962

Obs per group:

min = 2

avg = 2.9

max = 4

Log likelihood = -5413.4207

 LR chi2(17) = 1770.89
 Prob > chi2 = 0.0000

> interval]		phinact	Coefficient	Std. err.	z	P> z	[95% conf.
		age	.1761131	.0059858	29.42	0.000	.1643811
>	.1878452	yedu	-.0400949	.0193493	-2.07	0.038	-.0780189
>	-.0021709	gender					
		Female	0	(omitted)			
		thinc					
	Below country average income		.1874996	.1259732	1.49	0.137	-.0594034
>	.4344026	Above country average income	.1437623	.1334624	1.08	0.281	-.1178191
>	.4053437	mst					
	Away from partner		-.1226939	.3383987	-0.36	0.717	-.7859431
>	.5405553	Never married	-.2117363	.5574305	-0.38	0.704	-1.30428
>	.8808074	Lost partner	.0300787	.1612926	0.19	0.852	-.2860489
>	.3462063	changed					
	No longer officially together		-.0384458	.1540652	-0.25	0.803	-.340408
>	.2635165	Newly officially together	-.2438509	.3543099	-0.69	0.491	-.9382856
>	.4505838	gali					
	Limited		.6927275	.0493872	14.03	0.000	.5959303
>	.7895247	cjs					
	Employed or self-employed		.5220355	.1022354	5.11	0.000	.3216579
>	.7224131	Unemployed	.5511421	.174407	3.16	0.002	.2093107
>	.8929735	Permanently sick	.7985226	.1227411	6.51	0.000	.5579545
>	1.039091	Homemaker	.1647436	.0975357	1.69	0.091	-.0264229
>	.3559101	Other	.5360843	.1401564	3.82	0.000	.2613828
>	.8107858	bmi	-.0061261	.0074079	-0.83	0.408	-.0206452
>	.008393	eurod	.1054005	.0090707	11.62	0.000	.0876222
>	.1231788	country					
	Germany	0	(omitted)				
	Sweden	0	(omitted)				
	Spain	0	(omitted)				
	Italy	0	(omitted)				
	France	0	(omitted)				
	Denmark	0	(omitted)				
	Greece	0	(omitted)				
	Switzerland	0	(omitted)				
	Belgium	0	(omitted)				
	Czechia	0	(omitted)				
	Poland	0	(omitted)				

126 estimates store FE

127 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) FE	(B) RE		
age	.1761131	.0955753	.0805378	.0055131
yedu	-.0400949	-.0288965	-.0111984	.018915
thinc				
1	.1874996	-.4534341	.6409336	.0813691
2	.1437623	-.6560687	.7998309	.0894295
mst				
1	-.1226939	.1015575	-.2242513	.3332869
2	-.2117363	.3300291	-.5417654	.5527878
3	.0300787	.090229	-.0601503	.1548489
changed				
0	-.0384458	-.2940148	.2555691	.1233099
2	-.2438509	.3953265	-.6391774	.2569752
1.gali	.6927275	1.417282	-.7245549	.0361624
cjs				
2	.5220355	.2222638	.2997717	.082562
3	.5511421	.5877893	-.0366472	.1283577
4	.7985226	1.690957	-.8924346	.0961576
5	.1647436	.1891099	-.0243662	.0815403
6	.5360843	.6676439	-.1315596	.1037688
bmi	-.0061261	.0477364	-.0538625	.0066881
eurod	.1054005	.192974	-.0875735	.0066949

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(17) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= **1113.01**
Prob > chi2 = **0.0000**
Click to Open File: [report.doc](#)

128

129 xtlogit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.eu
> rod i.country, fe or baselevels
note: multiple positive outcomes within groups encountered.
note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.
note: **2.gender** omitted because of no within-group variance.
note: **12.country** omitted because of no within-group variance.
note: **13.country** omitted because of no within-group variance.
note: **15.country** omitted because of no within-group variance.
note: **16.country** omitted because of no within-group variance.
note: **17.country** omitted because of no within-group variance.
note: **18.country** omitted because of no within-group variance.
note: **19.country** omitted because of no within-group variance.
note: **20.country** omitted because of no within-group variance.
note: **23.country** omitted because of no within-group variance.
note: **28.country** omitted because of no within-group variance.
note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5822.4756**
Iteration 1: Log likelihood = **-5413.7224**
Iteration 2: Log likelihood = **-5411.9886**
Iteration 3: Log likelihood = **-5411.9885**

Conditional fixed-effects logistic regression
Group variable: **id**

Number of obs = **17,262**
Number of groups = **5,962**

Obs per group:
 min = 2
 avg = 2.9
 max = 4

Log likelihood = -5411.9885

LR chi2(19) = 1773.76
 Prob > chi2 = 0.0000

		phinact	Odds ratio	Std. err.	z	P> z	[95% conf.]
> interval]							
	age		1.192475	.0071401	29.40	0.000	1.178562
> 1.206552	yedu		.9604761	.0185742	-2.09	0.037	.9247526
> .9975797							
	gender						
	Male	1	(base)				
	Female	1	(omitted)				
	thinc						
	No income	1	(base)				
Below country average income			1.200751	.1511905	1.45	0.146	.9381566
> 1.536847							
Above country average income			1.150029	.1534278	1.05	0.295	.8854179
> 1.493721							
	mst						
	Officially with partner	1	(base)				
	Away from partner		.8476568	.2900945	-0.48	0.629	.4334235
> 1.657783							
> 2.349953	Never married		.7825463	.4390316	-0.44	0.662	.2605918
> 1.424947	Lost partner		1.03798	.1678054	0.23	0.818	.7560996
	gali						
	Not limited	1	(base)				
> 2.201355	Limited		1.998207	.0987121	14.01	0.000	1.813806
	cjs						
	Retired	1	(base)				
	Employed or self-employed		1.686989	.172557	5.11	0.000	1.380528
> 2.06148							
> 2.441024	Unemployed		1.733926	.3025838	3.15	0.002	1.231655
> 2.821606	Permanently sick		2.218185	.272318	6.49	0.000	1.743809
> 1.425342	Homemaker		1.177361	.1148165	1.67	0.094	.9725242
> 2.252756	Other		1.711756	.2398554	3.84	0.000	1.300677
> 1.008326	bmi		.9937933	.0073609	-0.84	0.401	.9794704
	changed						
No longer officially together			1.305515	.3142663	1.11	0.268	.8144767
> 2.092593							
	All other situations	1	(base)				
Newly officially together			.6434043	.3517823	-0.81	0.420	.2203374
> 1.878796							
	eurod		1.113732	.0102544	11.70	0.000	1.093814
> 1.134013							
	changed#c.eurod						
No longer officially together			.9380614	.0366094	-1.64	0.101	.8689839

```

> 1.01263
Newly officially together | 1.055049 .1270821 0.44 0.656 .8331904
> 1.335983

```

```

country
Austria 1 (base)
Germany 1 (omitted)
Sweden 1 (omitted)
Spain 1 (omitted)
Italy 1 (omitted)
France 1 (omitted)
Denmark 1 (omitted)
Greece 1 (omitted)
Switzerland 1 (omitted)
Belgium 1 (omitted)
Czechia 1 (omitted)
Poland 1 (omitted)

```

```

130 outreg2 using results.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(**,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

```

```
131
```

```

132 xtlogit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.eu
> rod i.country, or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33239.682
Iteration 2: Log pseudolikelihood = -31474.473
Iteration 3: Log pseudolikelihood = -31437.471
Iteration 4: Log pseudolikelihood = -31437.34
Iteration 5: Log pseudolikelihood = -31437.34

```

Fitting full model:

```

tau = 0.0 Log pseudolikelihood = -31437.34
tau = 0.1 Log pseudolikelihood = -31263.228
tau = 0.2 Log pseudolikelihood = -31097.95
tau = 0.3 Log pseudolikelihood = -30953.553
tau = 0.4 Log pseudolikelihood = -30850.367
tau = 0.5 Log pseudolikelihood = -30823.935
tau = 0.6 Log pseudolikelihood = -30940.38

```

```

Iteration 0: Log pseudolikelihood = -30823.935
Iteration 1: Log pseudolikelihood = -30271.52
Iteration 2: Log pseudolikelihood = -30234.658
Iteration 3: Log pseudolikelihood = -30234.264
Iteration 4: Log pseudolikelihood = -30234.264

```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Random effects $u_i \sim \text{Gaussian}$

Integration method: **mvaghermite**

Log pseudolikelihood = **-30234.264**

Number of obs = **110,045**
Number of groups = **48,341**

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration pts. = **12**

Wald chi2(31) = **6751.53**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **48,341** clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.100284	.0026932	39.04	0.000	1.095018
>	1.105575	yedu	.9715408	.0040135	-6.99	0.000	.9637063
>	.9794391						
		gender					
		Male	1 (base)				
		Female	1.018452	.0366002	0.51	0.611	.9491848
>	1.092773						
		thinc					
		No income	1 (base)				
		Below country average income	.6352668	.0669706	-4.30	0.000	.5166797
>	.7810716						
		Above country average income	.5188725	.0562294	-6.05	0.000	.4195824
>	.6416587						
		mst					
		Officially with partner	1 (base)				
		Away from partner	1.106227	.0666551	1.68	0.094	.9830044
>	1.244895						
		Never married	1.391031	.1000805	4.59	0.000	1.208079
>	1.601689						
		Lost partner	1.094461	.0489174	2.02	0.043	1.002665
>	1.194663						
		gali					
		Not limited	1 (base)				
		Limited	4.12457	.1407755	41.52	0.000	3.857682
>	4.409923						
		cjs					
		Retired	1 (base)				
		Employed or self-employed	1.248769	.077385	3.58	0.000	1.105946
>	1.410036						
		Unemployed	1.799226	.22072	4.79	0.000	1.414701
>	2.288268						
		Permanently sick	5.423054	.431635	21.24	0.000	4.63975
>	6.338599						
		Homemaker	1.208419	.0664931	3.44	0.001	1.084877
>	1.346031						
		Other	1.950084	.1936673	6.72	0.000	1.605161
>	2.369126						
		bmi	1.048883	.0036886	13.57	0.000	1.041678
>	1.056137						
		changed					
		No longer officially together	.8223472	.1413465	-1.14	0.255	.5871502
>	1.151758						
		All other situations	1 (base)				
		Newly officially together	1.648611	.6431615	1.28	0.200	.7674404
>	3.541536						
		eurod	1.213761	.0078724	29.87	0.000	1.198429
>	1.229289						
		changed#c.eurod					
		No longer officially together	.9785177	.0316836	-0.67	0.502	.9183483
>	1.042629						
		Newly officially together	.9661077	.098555	-0.34	0.735	.7910289
>	1.179937						

	country		1	(base)			
	Austria						
>	Germany		.7929682	.0703206	-2.62	0.009	.6664554
>	Sweden		.3833123	.0374435	-9.82	0.000	.316522
>	Spain		2.421917	.1936223	11.06	0.000	2.070661
>	Italy		6.268478	.492744	23.35	0.000	5.373436
>	France		1.403596	.1214058	3.92	0.000	1.184722
>	Denmark		.6467573	.0633372	-4.45	0.000	.5338052
>	Greece		1.260178	.1060597	2.75	0.006	1.068545
>	Switzerland		.4490852	.0487063	-7.38	0.000	.3630866
>	Belgium		1.504337	.1217109	5.05	0.000	1.283741
>	Czechia		1.307848	.1086836	3.23	0.001	1.111275
>	Poland		2.290976	.2193195	8.66	0.000	1.899037
>	_cons		5.14e-06	1.40e-06	-44.74	0.000	3.01e-06
>	/lnsig2u		1.021688	.0381782			.9468602
>	sigma_u		1.666697	.0318158			1.605492
>	rho		.4578107	.0094766			.439304

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

133 * /// Mundlak_Gal_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Gi_* Mundlak_Ma
> r_* ///
>
134 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

135 asdoc, text(xtologitLim) save(report.doc) append
Click to Open File: report.doc

136 *asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_C_changed No Mundl
> ak_C_changed All Mundlak_C_changed Newly_Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Emp
> loyed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemak
> er Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_Oth_th
> inc Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_eurod Mun
> dlak_Gi_1 Mundlak_Gi_2 Mundlak_Gi_3 Mundlak_Mar_mst_Officially Mundlak_Mar_mst_Away
> Mundlak_Mar_mst_Never Mundlak_Mar_mst_Lost, save(report.doc) append

```

137

```
138 logit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.euro
> d i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84670512
Iteration 2: Log pseudolikelihood = -81005214
Iteration 3: Log pseudolikelihood = -80947926
Iteration 4: Log pseudolikelihood = -80947802
Iteration 5: Log pseudolikelihood = -80947802
```

Logistic regression

Number of obs = 110,007

Wald chi2(34) = 4755.92

Prob > chi2 = 0.0000

Pseudo R2 = 0.1992

Log pseudolikelihood = -80947802

(Std. err. adjusted for 48,310 clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.059557	.0033179	18.47	0.000	1.053074
>	1.06608	yedu	.9777868	.00535	-4.11	0.000	.967357
>	.9883292	gender					
		Male	1 (base)				
		Female	.9882358	.0445793	-0.26	0.793	.9046132
>	1.079588	thinc					
		No income	1 (base)				
		Below country average income	.5367557	.0798976	-4.18	0.000	.4009341
>	.7185887	Above country average income	.4376391	.0663829	-5.45	0.000	.3250891
>	.5891554	mst					
		Officially with partner	1 (base)				
		Away from partner	1.262653	.1087642	2.71	0.007	1.066503
>	1.494879	Never married	1.213222	.1070236	2.19	0.028	1.020591
>	1.44221	Lost partner	1.108644	.0593565	1.93	0.054	.9982028
>	1.231303	gali					
		Not limited	1 (base)				
		Limited	2.902283	.1255669	24.63	0.000	2.666322
>	3.159125	cjs					
		Retired	1 (base)				
		Employed or self-employed	1.016471	.084854	0.20	0.845	.8630531
>	1.197161	Unemployed	1.74188	.2782515	3.47	0.001	1.273637
>	2.38227	Permanently sick	3.573798	.3696004	12.32	0.000	2.918094
>	4.376842	Homemaker	1.154182	.0776852	2.13	0.033	1.011538
>	1.316942	Other	1.424528	.1594382	3.16	0.002	1.143936
>	1.773946	bmi	1.038227	.0046449	8.39	0.000	1.029163
>	1.047371						

changed						
No longer officially together		1.048389	.2474198	0.20	0.841	.6601445
> 1.664969						
All other situations		1	(base)			
Newly officially together		1.348295	.5580478	0.72	0.470	.5990735
> 3.03452						
eurod						
> 1.162988		1.145244	.0089839	17.29	0.000	1.12777
changed#c.eurod						
No longer officially together		.9693542	.0373145	-0.81	0.419	.8989099
> 1.045319						
Newly officially together		.9678382	.0899348	-0.35	0.725	.8066893
> 1.161179						
country						
Austria	1	(base)				
> 1.114768		.9438142	.0801643	-0.68	0.496	.7990766
Germany						
> .5959952		.4871687	.0501152	-6.99	0.000	.3982135
Sweden						
> 2.625863		2.194203	.2010543	8.58	0.000	1.833502
Spain						
> 4.965928		4.282793	.3233891	19.26	0.000	3.693634
Italy						
> 1.541045		1.318101	.1050929	3.46	0.001	1.12741
France						
> .9368148		.7815697	.0722492	-2.67	0.008	.6520511
Denmark						
> 1.512636		1.297809	.101427	3.34	0.001	1.113493
Greece						
> .7304135		.5959958	.061844	-4.99	0.000	.4863149
Switzerland						
> 1.534449		1.315297	.1034196	3.49	0.000	1.127445
Belgium						
> 1.462897		1.192624	.1242925	1.69	0.091	.9722848
Czechia						
> 2.130106		1.792868	.1576611	6.64	0.000	1.509022
Poland						
Qyear						
2015	1	(base)				
> .8714188		.7973703	.0361279	-5.00	0.000	.7296141
2017						
> 1.057632		.9640212	.0455826	-0.77	0.438	.8786957
2020						
> 1.079519		.9908455	.0433312	-0.21	0.833	.9094557
2022						
> .0007963		.0004148	.000138	-23.40	0.000	.000216
_cons						

Note: **_cons** estimates baseline odds.

139 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
 > title(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
 results.xls
 dir : seeout

140

141

```
142 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.changed c.
> eurod i.country
```

Fitting comparison model:

```
Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33239.64
Iteration 2: Log likelihood = -31480.962
Iteration 3: Log likelihood = -31444.267
Iteration 4: Log likelihood = -31444.137
Iteration 5: Log likelihood = -31444.137
```

Fitting full model:

```
tau = 0.0 Log likelihood = -31444.137
tau = 0.1 Log likelihood = -31269.897
tau = 0.2 Log likelihood = -31104.505
tau = 0.3 Log likelihood = -30960.02
tau = 0.4 Log likelihood = -30856.786
tau = 0.5 Log likelihood = -30830.367
tau = 0.6 Log likelihood = -30946.906
```

```
Iteration 0: Log likelihood = -30830.367
Iteration 1: Log likelihood = -30277.666
Iteration 2: Log likelihood = -30240.897
Iteration 3: Log likelihood = -30240.505
Iteration 4: Log likelihood = -30240.505
```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**Integration pts. = **12**Log likelihood = **-30240.505**

Wald chi2(27) = **7020.04**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0944435	.0022743	41.53	0.000	.0899859
> .098901	yedu	-.0286127	.0040701	-7.03	0.000	-.0365899
> -.0206356	gender Female	.0069056	.0353916	0.20	0.845	-.0624607
> .0762719	thinc Below country average income	-.4543356	.0961547	-4.73	0.000	-.6427953
> -.2658758	Above country average income	-.6550089	.0992557	-6.60	0.000	-.8495465
> -.4604713	single Yes	.1261672	.0368881	3.42	0.001	.0538679
> .1984665	gali Limited	1.418308	.033633	42.17	0.000	1.352388
> 1.484227						

	cjs					
>	Employed or self-employed	.2220563	.0602765	3.68	0.000	.1039166
>	.3401961					
>	.8242896	Unemployed	.5928962	.11806	5.02	0.000
>	1.848584	Permanently sick	1.699129	.0762539	22.28	0.000
>	.2876202	Homemaker	.1828976	.0534309	3.42	0.001
>	.8443877	Other	.6599053	.0941254	7.01	0.000
>	.0537099	bmi	.0474716	.0031828	14.91	0.000
>	-.1382038	changed	-.3149643	.0901855	-3.49	0.000
>	.8359426	No longer officially together	.3578835	.2439122	1.47	0.142
>	.2046442	Newly officially together	.1926466	.0061213	31.47	0.000
>	-.062946	eurod	-.2358335	.0882095	-2.67	0.008
>	-.7658648	country	-.9579178	.0979881	-9.78	0.000
>	1.040105	Germany	.8820364	.0806485	10.94	0.000
>	1.984186	Sweden	1.830233	.078549	23.30	0.000
>	.5094096	Spain	.3400466	.0864113	3.94	0.000
>	-.2409093	Italy	-.4345574	.0988019	-4.40	0.000
>	.3870298	France	.2233083	.0835329	2.67	0.008
>	-.5886102	Denmark	-.7993548	.1075248	-7.43	0.000
>	.5625554	Greece	.4032893	.0812597	4.96	0.000
>	.4202294	Switzerland	.2568385	.0833642	3.08	0.002
>	1.003433	Belgium	.8151587	.09606	8.49	0.000
>	-11.58382	Czechia	-12.07633	.2512854	-48.06	0.000
>		Poland				
>		_cons				
>	1.094728	/lnsig2u	1.021233	.0374984		.9477377
>	1.728691	sigma_u	1.666318	.0312421		1.606196
>	.4759887	rho	.4576978	.0093075		.4395201

LR test of rho=0: chibar2(01) = 2407.26

Prob >= chibar2 = 0.000

143 estimates store RE

144 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.changed c.
> eurod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **19.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5826.2722**

Iteration 1: Log likelihood = **-5415.1119**

Iteration 2: Log likelihood = **-5413.4211**

Iteration 3: Log likelihood = **-5413.421**

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = **17,262**

Number of groups = **5,962**

Obs per group:

min = **2**

avg = **2.9**

max = **4**

Log likelihood = **-5413.421**

LR chi2(15) = **1770.89**

Prob > chi2 = **0.0000**

		phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]							
	age		.1771974	.0060019	29.52	0.000	.1654339
> .188961	yedu		-.039822	.0193161	-2.06	0.039	-.0776809
> -.0019631							
	gender						
	Female		0	(omitted)			
	thinc						
	Below country average income		.1864671	.1259642	1.48	0.139	-.0604182
> .4333524	Above country average income		.1387266	.1335789	1.04	0.299	-.1230831
> .4005364							
	single						
	Yes		-.0610691	.1106247	-0.55	0.581	-.2778896
> .1557513							
	gali						
	Limited		.6926735	.0493889	14.02	0.000	.595873
> .789474							
	cjs						
	Employed or self-employed		.5234409	.1021465	5.12	0.000	.3232375
> .7236443	Unemployed		.5549879	.174439	3.18	0.001	.2130938
> .896882	Permanently sick		.8016875	.1227456	6.53	0.000	.5611105
> 1.042264	Homemaker		.1631517	.097527	1.67	0.094	-.0279977

```

> .3543011
> .8135211      Other | .5388196 .1401564 3.84 0.000 .2641182
> .0082688      bmi | -.0062586 .0074121 -0.84 0.398 -.0207859
> .0082688      changed
No longer officially together | .0165795 .1303156 0.13 0.899 -.2388343
> .2719933      Newly officially together | -.1837052 .3042401 -0.60 0.546 -.7800048
> .4125943      eurod | .1054186 .0090695 11.62 0.000 .0876426
> .1231945
country
Germany      0 (omitted)
Sweden      0 (omitted)
Spain      0 (omitted)
Italy      0 (omitted)
France      0 (omitted)
Denmark      0 (omitted)
Greece      0 (omitted)
Switzerland 0 (omitted)
Belgium      0 (omitted)
Czechia      0 (omitted)
Poland      0 (omitted)

```

145 estimates store FE

146 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	$\sqrt{\text{diag}(V_b - V_B)}$ Std. err.
age	.1771974	.0944435	.082754	.0055543
yedu	-.039822	-.0286127	-.0112092	.0188824
thinc				
1	.1864671	-.4543356	.6408027	.0813711
2	.1387266	-.6550089	.7937356	.0893959
1.single	-.0610691	.1261672	-.1872363	.1042933
1.gali	.6926735	1.418308	-.7256342	.0361674
cjs				
2	.5234409	.2220563	.3013846	.082466
3	.5549879	.5928962	-.0379083	.1284165
4	.8016875	1.699129	-.8974414	.0961864
5	.1631517	.1828976	-.0197459	.0815883
6	.5388196	.6599053	-.1210857	.1038471
bmi	-.0062586	.0474716	-.0537302	.0066939
changed				
0	.0165795	-.3149643	.3315438	.0940676
2	-.1837052	.3578835	-.5415888	.1818485
eurod	.1054186	.1926466	-.0872281	.0066922

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

```

chi2(15) = (b-B)'[(V_b-V_B)^(-1)](b-B)
          = 1117.62
Prob > chi2 = 0.0000
Click to Open File: report.doc

```

147

148

149 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.schanged##

> c.eurod i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.Iteration 0: Log likelihood = **-5822.8804**Iteration 1: Log likelihood = **-5411.5765**Iteration 2: Log likelihood = **-5409.7982**Iteration 3: Log likelihood = **-5409.7981**

Conditional fixed-effects logistic regression

Group variable: **id**Number of obs = **17,262**Number of groups = **5,962**

Obs per group:

min = **2**avg = **2.9**max = **4**Log likelihood = **-5409.7981**LR chi2(17) = **1778.14**Prob > chi2 = **0.0000**

> interval]		phinact	Odds ratio	Std. err.	z	P> z	[95% conf.
		age	1.193982	.0072111	29.36	0.000	1.179932
> 1.208199		yedu	.9611865	.0185572	-2.05	0.040	.9254946
> .9982549		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		thinc					
		No income	1	(base)			
Below country average income			1.199121	.1509844	1.44	0.149	.9368845
> 1.534758		Above country average income	1.145213	.1529077	1.02	0.310	.8815247
> 1.487777		single					
		No	1	(base)			
> 1.354112		Yes	1.027117	.1448418	0.19	0.850	.7790858
		gali					
		Not limited	1	(base)			
> 2.199543		Limited	1.996492	.0986631	13.99	0.000	1.812185
		cjs					
		Retired	1	(base)			
> 2.075888		Employed or self-employed	1.699259	.1735673	5.19	0.000	1.390962

	Unemployed	1.744534	.3045186	3.19	0.001	1.239074
> 2.456188						
	Permanently sick	2.217455	.2724094	6.48	0.000	1.742956
> 2.821129						
	Homemaker	1.177278	.1148587	1.67	0.094	.9723742
> 1.425361						
	Other	1.719782	.2410161	3.87	0.000	1.306722
> 2.263412						
	bmi	.9939817	.0073702	-0.81	0.416	.9796408
> 1.008533						
	changed					
	Hooked up	.4582858	.2150017	-1.66	0.096	.1827262
> 1.149402						
	Unchanged	1	(base)			
	Back single	1.130955	.2278444	0.61	0.541	.7620101
> 1.678534						
	eurod	1.114476	.0103228	11.70	0.000	1.094426
> 1.134893						
	schanged#c.eurod					
	Hooked up	1.197643	.1373797	1.57	0.116	.9565052
> 1.499572						
	Back single	.9449816	.0309743	-1.73	0.084	.8861821
> 1.007683						
	country					
	Austria	1	(base)			
	Germany	1	(omitted)			
	Sweden	1	(omitted)			
	Spain	1	(omitted)			
	Italy	1	(omitted)			
	France	1	(omitted)			
	Denmark	1	(omitted)			
	Greece	1	(omitted)			
	Switzerland	1	(omitted)			
	Belgium	1	(omitted)			
	Czechia	1	(omitted)			
	Poland	1	(omitted)			

```

150 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

```

```

151
152 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.schanged##
> c.eurod i.country, or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33241.208
Iteration 2: Log pseudolikelihood = -31482.129
Iteration 3: Log pseudolikelihood = -31445.367
Iteration 4: Log pseudolikelihood = -31445.237
Iteration 5: Log pseudolikelihood = -31445.237

```

Fitting full model:

```

tau = 0.0 Log pseudolikelihood = -31445.237
tau = 0.1 Log pseudolikelihood = -31270.842
tau = 0.2 Log pseudolikelihood = -31105.251
tau = 0.3 Log pseudolikelihood = -30960.513
tau = 0.4 Log pseudolikelihood = -30856.967
tau = 0.5 Log pseudolikelihood = -30830.169
tau = 0.6 Log pseudolikelihood = -30946.262

```

Iteration 0: Log pseudolikelihood = **-30830.169**
 Iteration 1: Log pseudolikelihood = **-30276.529**
 Iteration 2: Log pseudolikelihood = **-30239.092**
 Iteration 3: Log pseudolikelihood = **-30238.683**
 Iteration 4: Log pseudolikelihood = **-30238.682**

Calculating robust standard errors ...

Random-effects logistic regression
 Group variable: **id**

Number of obs = **110,045**
 Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
 min = **1**
 avg = **2.3**
 max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-30238.682**

Wald chi2(29) = **6746.55**
 Prob > chi2 = **0.0000**

(Std. err. adjusted for **48,341** clusters)

> rs in **id**)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.]
> interval]						
	age	1.099084	.0026407	39.32	0.000	1.093921
> 1.104272	yedu	.9718298	.0040098	-6.93	0.000	.9640024
> .9797207						
	gender					
	Male	1 (base)				
	Female	1.004234	.0361913	0.12	0.907	.9357479
> 1.077733						
	thinc					
	No income	1 (base)				
	Below country average income	.6333401	.0667401	-4.33	0.000	.5151564
> .7786367						
	Above country average income	.5200934	.0564884	-6.02	0.000	.420369
> .6434755						
	single					
	No	1 (base)				
	Yes	1.141271	.0437003	3.45	0.001	1.058755
> 1.230218						
	gali					
	Not limited	1 (base)				
	Limited	4.129521	.1410353	41.52	0.000	3.862145
> 4.415406						
	cjs					
	Retired	1 (base)				
	Employed or self-employed	1.249589	.0774523	3.59	0.000	1.106643
> 1.410998						
	Unemployed	1.810656	.2219844	4.84	0.000	1.4239
> 2.302461						
	Permanently sick	5.467262	.4350562	21.35	0.000	4.677736
> 6.390047						
	Homemaker	1.203228	.0661259	3.37	0.001	1.08036
> 1.34007						
	Other	1.936107	.1921302	6.66	0.000	1.593896
> 2.35179						
	bmi	1.048611	.0036855	13.51	0.000	1.041412
> 1.055859						

> 1.267423	schanged Hooked up	.5349998	.2354246	-1.42	0.155	.2258321
> 1.25873	Unchanged Back single	1 .9478389	(base) .1371846	-0.37	0.711	.7137344
> 1.230153	eurod	1.214528	.007921	29.80	0.000	1.199102
> 1.371229	schanged#c.eurod Hooked up	1.14277	.1062631	1.44	0.151	.9523749
> 1.006251	Back single	.9517157	.0270567	-1.74	0.082	.900136
> .9413114	country Austria	1 .7910965	(base) .0701724	-2.64	0.008	.6648529
> .4645321	Germany	.3834927	.0375106	-9.80	0.000	.316591
> 2.825892	Sweden	2.416383	.1930087	11.05	0.000	2.066218
> 7.278386	Spain	6.241077	.4896037	23.34	0.000	5.351604
> 1.662711	Italy	1.403337	.1214311	3.92	0.000	1.184424
> .784974	France	.6477158	.0635165	-4.43	0.000	.5344582
> 1.473376	Denmark	1.249568	.1050413	2.65	0.008	1.059756
> .5563213	Greece	.4497445	.0487996	-7.36	0.000	.363585
> 1.757109	Switzerland	1.499313	.1213721	5.00	0.000	1.27934
> 1.521264	Belgium	1.292956	.1072711	3.10	0.002	1.098913
> 2.727947	Czechia	2.262241	.2160632	8.55	0.000	1.876039
> 9.56e-06	Poland	5.66e-06	1.51e-06	-45.12	0.000	3.35e-06
> 1.098657	_cons	1.023961	.0381107			.9492654
> 1.732089	/lnsig2u	1.668593	.0317956			1.607424
> .4769686	sigma_u	.4583749	.0094616			.4398965
	rho					

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

153 * /// Mundlak_Gal_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_sGi_* Mundlak_
> Sin_* ///
>

```
154 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

```
155 asdoc, text(xtologitLim) save(report.doc) append
Click to Open File: report.doc
```

```
156 *asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_sC_schanged_Hooked
> Mundlak_sC_schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired Mund
> lak_Oth_cjs_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_
> Oth_cjs_Homemaker Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below
> Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_
> OthC_eurod Mundlak_sGi_1 Mundlak_sGi_2 Mundlak_sGi_3 Mundlak_Sin_single_No Mundlak_
> Sin_single_Yes, save(report.doc) append
```

```
157
158 logit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ib1.schanged#c.
> eurod i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84642684
Iteration 2: Log pseudolikelihood = -80978284
Iteration 3: Log pseudolikelihood = -80920915
Iteration 4: Log pseudolikelihood = -80920786
Iteration 5: Log pseudolikelihood = -80920786
```

Logistic regression

```
Number of obs = 110,007
Wald chi2(32) = 4747.55
Prob > chi2 = 0.0000
Pseudo R2 = 0.1995
```

Log pseudolikelihood = -80920786

(Std. err. adjusted for 48,310 clusters)

> rs in id)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.057707	.0032545	18.23	0.000	1.051347
> 1.064105						
	yedu	.9779381	.0053284	-4.09	0.000	.9675503
> .9884375						
	gender					
	Male	1 (base)				
	Female	.9714813	.0439564	-0.64	0.523	.8890379
> 1.06157						
	thinc					
	No income	1 (base)				
	Below country average income	.5355138	.0793087	-4.22	0.000	.4005983
> .7158669						
	Above country average income	.4421286	.0665224	-5.42	0.000	.3292134
> .5937721						
	single					
	No	1 (base)				
	Yes	1.196428	.0592897	3.62	0.000	1.085687
> 1.318464						
	gali					
	Not limited	1 (base)				
	Limited	2.911047	.1258241	24.72	0.000	2.674593
> 3.168405						
	cjs					
	Retired	1 (base)				
	Employed or self-employed	1.015249	.0850707	0.18	0.857	.8614849
> 1.196457						
	Unemployed	1.750519	.277252	3.54	0.000	1.283368

>	2.387715	Permanently sick	3.583356	.3684595	12.41	0.000	2.929308
>	4.383439						
>	1.309347						
>	1.7384						
>	1.047163	bmi	1.038067	.0046206	8.39	0.000	1.029051
>	1.556769	schanged Hooked up	.6028327	.2918035	-1.05	0.296	.2334369
>	1.43184	Unchanged Back single	1 .9803691	(base) .1894679	-0.10	0.918	.6712509
>	1.163892	eurod	1.146075	.0090208	17.32	0.000	1.12853
>	1.188984	schanged#c.eurod Hooked up	.9667905	.102044	-0.32	0.749	.7861196
>	1.024871	Back single	.9589651	.0325211	-1.24	0.217	.8972971
>	1.112466	country Austria Germany	1 .941933	(base) .0799698	-0.70	0.481	.7975415
>	.5976023	Sweden	.4882099	.0503613	-6.95	0.000	.3988421
>	2.592304	Spain	2.167038	.1981174	8.46	0.000	1.811536
>	4.888928	Italy	4.222204	.3158434	19.25	0.000	3.646404
>	1.534022	France	1.312142	.104593	3.41	0.001	1.122355
>	.9397223	Denmark	.7838422	.0725374	-2.63	0.008	.6538194
>	1.491553	Greece	1.280307	.0997599	3.17	0.002	1.098979
>	.7328927	Switzerland	.5980315	.0620487	-4.96	0.000	.4879864
>	1.537887	Belgium	1.317725	.1038753	3.50	0.000	1.129081
>	1.451254	Czechia	1.183223	.1232662	1.61	0.106	.9646948
>	2.078159	Poland	1.751687	.1527419	6.43	0.000	1.476503
>	.8809112	Qyear 2015 2017	1 .8061496	(base) .0364779	-4.76	0.000	.7377329
>	1.072872	2020	.9776645	.0463539	-0.48	0.634	.8909061
>	1.09892	2022	1.008398	.0442286	0.19	0.849	.9253334
>	.000892	_cons	.0004715	.0001534	-23.55	0.000	.0002492

Note: **_cons** estimates baseline odds.

```

159 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

```

```
160
```

```
161
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162
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163
```

```
164
```

```

165 xtlogit phinact age yedu i.gender i.thinc i.mst ib1.changed i.sphus i.cjs c.bmi c.eu
> rod i.country

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33341.273
Iteration 2: Log likelihood = -30402.042
Iteration 3: Log likelihood = -30159.285
Iteration 4: Log likelihood = -30158.011
Iteration 5: Log likelihood = -30158.011

```

Fitting full model:

```

tau = 0.0 Log likelihood = -30158.011
tau = 0.1 Log likelihood = -29999.182
tau = 0.2 Log likelihood = -29850.649
tau = 0.3 Log likelihood = -29724.821
tau = 0.4 Log likelihood = -29642.732
tau = 0.5 Log likelihood = -29641.353
tau = 0.6 Log likelihood = -29789.676

```

```

Iteration 0: Log likelihood = -29641.352
Iteration 1: Log likelihood = -29115.796
Iteration 2: Log likelihood = -29112.761
Iteration 3: Log likelihood = -29112.755
Iteration 4: Log likelihood = -29112.755

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-29112.755**

Wald chi2(32) = **7815.66**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0925454	.0022957	40.31	0.000	.0880459
> .0970448	yedu	-.0181854	.0040488	-4.49	0.000	-.0261208
> -.0102499	gender					
	Female	.1283035	.0352808	3.64	0.000	.0591545
> .1974526	thinc					
	Below country average income	-.570086	.0950744	-6.00	0.000	-.7564284
> -.3837435	Above country average income	-.7219872	.0978989	-7.37	0.000	-.9138655
> -.5301088						

		mst					
>	.1826511	Away from partner	.0683493	.0583183	1.17	0.241	-.0459525
>	.4525463	Never married	.3132433	.0710743	4.41	0.000	.1739403
>	.1978105	Lost partner	.1101811	.0447097	2.46	0.014	.0225517
		changed					
No		longer officially together					
>	-.0609114		-.2435271	.093173	-2.61	0.009	-.4261427
>	1.035928	Newly officially together	.5536561	.2460617	2.25	0.024	.0713839
		sphus					
>	.1552382	Very good	-.0376238	.0984008	-0.38	0.702	-.2304859
>	.5275672	Good	.3490693	.0910721	3.83	0.000	.1705713
>	1.506107	Fair	1.326277	.0917519	14.46	0.000	1.146446
>	3.245931	Poor	3.054535	.0976529	31.28	0.000	2.863139
		cjs					
>	.4167737	Employed or self-employed	.2988579	.0601622	4.97	0.000	.1809421
>	.8372417	Unemployed	.6068024	.1175732	5.16	0.000	.3763631
>	1.587017	Permanently sick	1.43696	.076561	18.77	0.000	1.286904
>	.3001407	Homemaker	.1960264	.0531205	3.69	0.000	.0919121
>	.8041234	Other	.6191805	.0943603	6.56	0.000	.4342376
		bmi					
>	.0550295		.0488215	.0031674	15.41	0.000	.0426135
>	.1284691	eurod	.1161214	.0063	18.43	0.000	.1037737
		country					
>	-.1637156	Germany	-.3371138	.0884701	-3.81	0.000	-.5105121
>	-.7134061	Sweden	-.9056884	.098105	-9.23	0.000	-1.097971
>	.7528846	Spain	.5956352	.0802307	7.42	0.000	.4383859
>	1.817849	Italy	1.665115	.0779266	21.37	0.000	1.512382
>	.4058242	France	.2368863	.0861944	2.75	0.006	.0679485
>	-.2001903	Denmark	-.3938354	.0988003	-3.99	0.000	-.5874804
>	.1106427	Greece	-.0519691	.0829668	-0.63	0.531	-.214581
>	-.4657538	Switzerland	-.675283	.1069046	-6.32	0.000	-.8848122
>	.7242679	Belgium	.565509	.0810009	6.98	0.000	.4067501
>	.3323202	Czechia	.168629	.0835175	2.02	0.043	.0049378
>	.6687777	Poland	.4804049	.0961103	5.00	0.000	.2920322
>	-11.37568	_cons	-11.89399	.2644476	-44.98	0.000	-12.4123
>	.994412	/lnsig2u	.9185522	.0387047			.8426924

	sigma_u	1.582928	.0306334	1.524012
> 1.644121	rho	.4323437	.009499	.4138299
> .4510484				

LR test of rho=0: chibar2(01) = 2090.51

Prob >= chibar2 = 0.000

166 estimates store RE

167 xtlogit phinact age yedu i.gender i.thinc i.mst ibl.changed i.sphus i.cjs c.bmi c.eu
> rod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5682.4207

Iteration 1: Log likelihood = -5285.1486

Iteration 2: Log likelihood = -5283.173

Iteration 3: Log likelihood = -5283.1724

Iteration 4: Log likelihood = -5283.1724

Conditional fixed-effects logistic regression
Group variable: **id**

Number of obs = 17,262

Number of groups = 5,962

Obs per group:

min = 2

avg = 2.9

max = 4

Log likelihood = -5283.1724

LR chi2(20) = 2031.39

Prob > chi2 = 0.0000

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.1723345	.0060571	28.45	0.000	.1604628
> .1842062	yedu	-.0419938	.0195573	-2.15	0.032	-.0803255
> -.0036621						
	gender					
	Female	0	(omitted)			
	thinc					
	Below country average income	.1548825	.1266892	1.22	0.222	-.0934238
> .4031889	Above country average income	.108924	.1343678	0.81	0.418	-.1544321
> .3722801						
	mst					
	Away from partner	-.0502066	.338236	-0.15	0.882	-.7131371
> .6127239	Never married	-.1146535	.5812424	-0.20	0.844	-1.253868
> 1.024561	Lost partner	.0473318	.1626894	0.29	0.771	-.2715335

>	.3661971					
No	changed					
	longer officially together	-.0562219	.1562495	-0.36	0.719	-.3624654
>	.2500215					
	Newly officially together	-.0941338	.3634057	-0.26	0.796	-.8063959
>	.6181283					
	sphus					
	Very good	-.2058793	.1434381	-1.44	0.151	-.4870128
>	.0752543					
	Good	-.0329768	.138807	-0.24	0.812	-.3050335
>	.2390798					
	Fair	.4373624	.1418099	3.08	0.002	.1594201
>	.7153047					
	Poor	1.375587	.1500294	9.17	0.000	1.081535
>	1.669639					
	cjs					
	Employed or self-employed	.5062779	.103154	4.91	0.000	.3040998
>	.708456					
	Unemployed	.5542604	.176214	3.15	0.002	.2088874
>	.8996334					
	Permanently sick	.7613586	.125886	6.05	0.000	.5146265
>	1.008091					
	Homemaker	.1339531	.0986252	1.36	0.174	-.0593488
>	.3272549					
	Other	.4709706	.1416474	3.32	0.001	.1933468
>	.7485944					
	bmi	-.0028905	.0075585	-0.38	0.702	-.0177049
>	.011924					
	eurod	.0756036	.0094236	8.02	0.000	.0571337
>	.0940736					
	country					
	Germany	0	(omitted)			
	Sweden	0	(omitted)			
	Spain	0	(omitted)			
	Italy	0	(omitted)			
	France	0	(omitted)			
	Denmark	0	(omitted)			
	Greece	0	(omitted)			
	Switzerland	0	(omitted)			
	Belgium	0	(omitted)			
	Czechia	0	(omitted)			
	Poland	0	(omitted)			

168 estimates store FE

169 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
age	.1723345	.0925454	.0797891	.0056052
yedu	-.0419938	-.0181854	-.0238084	.0191337
thinc				
1	.1548825	-.570086	.7249685	.0837318
2	.108924	-.7219872	.8309112	.0920354
mst				
1	-.0502066	.0683493	-.1185559	.3331705
2	-.1146535	.3132433	-.4278968	.5768805
3	.0473318	.1101811	-.0628493	.1564253
changed				
0	-.0562219	-.2435271	.1873051	.1254301
2	-.0941338	.5536561	-.6477899	.2674272
sphus				
2	-.2058793	-.0376238	-.1682554	.1043636

3	-.0329768	.3490693	-.3820461	.1047533
4	.4373624	1.326277	-.8889143	.1081279
5	1.375587	3.054535	-1.678948	.113898
cjs				
2	.5062779	.2988579	.20742	.0837929
3	.5542604	.6068024	-.052542	.1312551
4	.7613586	1.43696	-.6756018	.0999285
5	.1339531	.1960264	-.0620733	.0830972
6	.4709706	.6191805	-.1482099	.1056414
bmi	-.0028905	.0488215	-.051712	.0068629
eurod	.0756036	.1161214	-.0405177	.0070082

b = Consistent under H0 and Ha; obtained from **xtlogit**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(20) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = 1180.10
 Prob > chi2 = 0.0000
 Click to Open File: [report.doc](#)

170

```
171 xtlogit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed##c.e
> urod i.country, fe or baselevels
note: multiple positive outcomes within groups encountered.
note: 42,379 groups (92,783 obs) omitted because of all positive or
      all negative outcomes.
note: 2.gender omitted because of no within-group variance.
note: 12.country omitted because of no within-group variance.
note: 13.country omitted because of no within-group variance.
note: 15.country omitted because of no within-group variance.
note: 16.country omitted because of no within-group variance.
note: 17.country omitted because of no within-group variance.
note: 18.country omitted because of no within-group variance.
note: 19.country omitted because of no within-group variance.
note: 20.country omitted because of no within-group variance.
note: 23.country omitted because of no within-group variance.
note: 28.country omitted because of no within-group variance.
note: 29.country omitted because of no within-group variance.
```

Iteration 0: Log likelihood = -5681.7469
 Iteration 1: Log likelihood = -5284.056
 Iteration 2: Log likelihood = -5282.055
 Iteration 3: Log likelihood = -5282.0544
 Iteration 4: Log likelihood = -5282.0544

Conditional fixed-effects logistic regression
 Group variable: id

Number of obs = 17,262
 Number of groups = 5,962

Obs per group:
 min = 2
 avg = 2.9
 max = 4

Log likelihood = -5282.0544
 LR chi2(22) = 2033.63
 Prob > chi2 = 0.0000

	phinact	Odds ratio	Std. err.	z	P> z	[95% conf.
> interval]						
	age	1.187998	.0071977	28.43	0.000	1.173974
> 1.202189	yedu	.9585801	.0187356	-2.16	0.030	.9225534
> .9960137						
	gender					
	Male	1	(base)			
	Female	1	(omitted)			

	thinc					
	No income	1	(base)			
	Below country average income	1.162598	.1472864	1.19	0.234	.9069709
>	1.490274					
	Above country average income	1.110737	.1492458	0.78	0.434	.8535686
>	1.445387					
	mst					
	Officially with partner	1	(base)			
	Away from partner	.9038505	.3101604	-0.29	0.768	.4613206
>	1.770885					
	Never married	.8497704	.4996188	-0.28	0.782	.2684364
>	2.690059					
	Lost partner	1.052871	.1716716	0.32	0.752	.7648675
>	1.449321					
	sphus					
	Excellent	1	(base)			
	Very good	.816106	.11707	-1.42	0.157	.616087
>	1.081063					
	Good	.9710693	.1348225	-0.21	0.833	.7397247
>	1.274766					
	Fair	1.553807	.2204464	3.11	0.002	1.17661
>	2.051924					
	Poor	3.966757	.5953088	9.18	0.000	2.955915
>	5.323278					
	cjs					
	Retired	1	(base)			
	Employed or self-employed	1.659968	.1713239	4.91	0.000	1.355963
>	2.03213					
	Unemployed	1.739401	.3065802	3.14	0.002	1.231318
>	2.457136					
	Permanently sick	2.138068	.2692534	6.03	0.000	1.670426
>	2.736629					
	Homemaker	1.142063	.1126234	1.35	0.178	.9413465
>	1.385576					
	Other	1.603559	.2271449	3.33	0.001	1.21482
>	2.116694					
	bmi	.9970378	.0075356	-0.39	0.695	.9823772
>	1.011917					
	changed					
	No longer officially together	1.224085	.3012522	0.82	0.411	.7556622
>	1.982877					
	All other situations	1	(base)			
	Newly officially together	.6756415	.3806081	-0.70	0.486	.2239805
>	2.038086					
	eurod	1.080515	.0103358	8.10	0.000	1.060446
>	1.100964					
	changed#c.eurod					
	No longer officially together	.947853	.0378779	-1.34	0.180	.8764467
>	1.025077					
	Newly officially together	1.087142	.1354772	0.67	0.503	.851553
>	1.38791					
	country					
	Austria	1	(base)			
	Germany	1	(omitted)			
	Sweden	1	(omitted)			
	Spain	1	(omitted)			
	Italy	1	(omitted)			
	France	1	(omitted)			
	Denmark	1	(omitted)			
	Greece	1	(omitted)			
	Switzerland	1	(omitted)			
	Belgium	1	(omitted)			
	Czechia	1	(omitted)			

Poland	1 (omitted)
--------	-------------

```
172 outreg2 using results2.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(**,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

173

```
174 xtlogit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed#c.e
> urod i.country, or baselevels vce(robust)
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33341.394
Iteration 2: Log pseudolikelihood = -30401.873
Iteration 3: Log pseudolikelihood = -30159.018
Iteration 4: Log pseudolikelihood = -30157.743
Iteration 5: Log pseudolikelihood = -30157.743
```

Fitting full model:

```
tau = 0.0 Log pseudolikelihood = -30157.743
tau = 0.1 Log pseudolikelihood = -29998.92
tau = 0.2 Log pseudolikelihood = -29850.394
tau = 0.3 Log pseudolikelihood = -29724.571
tau = 0.4 Log pseudolikelihood = -29642.485
tau = 0.5 Log pseudolikelihood = -29641.104
tau = 0.6 Log pseudolikelihood = -29789.416
```

```
Iteration 0: Log pseudolikelihood = -29641.104
Iteration 1: Log pseudolikelihood = -29115.543
Iteration 2: Log pseudolikelihood = -29112.503
Iteration 3: Log pseudolikelihood = -29112.497
Iteration 4: Log pseudolikelihood = -29112.497
```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Number of obs = 110,045
Number of groups = 48,341

Random effects u_i ~ Gaussian

Obs per group:
min = 1
avg = 2.3
max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = **-29112.497**

Wald chi2(34) = 7584.17
Prob > chi2 = 0.0000

(Std. err. adjusted for 48,341 clust

> ers in **id**)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.096952	.0026466	38.35	0.000	1.091777
> 1.102151	yedu	.9819997	.0040293	-4.43	0.000	.9741341
> .9899288						
	gender					
	Male	1 (base)				
> 1.218584	Female	1.136647	.0403669	3.61	0.000	1.06022

	thinc					
	No income	1	(base)			
	Below country average income	.5653185	.0583587	-5.53	0.000	.4617664
>	.6920924					
	Above country average income	.4857647	.0514938	-6.81	0.000	.3946334
>	.5979408					
	mst					
	Officially with partner	1	(base)			
	Away from partner	1.070071	.0644638	1.12	0.261	.9508986
>	1.20418					
	Never married	1.367878	.0971987	4.41	0.000	1.190043
>	1.572288					
	Lost partner	1.116527	.0496922	2.48	0.013	1.023259
>	1.218296					
	sphus					
	Excellent	1	(base)			
	Very good	.9628823	.0942515	-0.39	0.699	.7947921
>	1.166522					
	Good	1.417676	.1303033	3.80	0.000	1.183969
>	1.697515					
	Fair	3.765921	.3484248	14.33	0.000	3.141359
>	4.514657					
	Poor	21.20334	2.100682	30.83	0.000	17.46115
>	25.74752					
	cjs					
	Retired	1	(base)			
	Employed or self-employed	1.348267	.0830286	4.85	0.000	1.194971
>	1.521228					
	Unemployed	1.834024	.2219212	5.01	0.000	1.446797
>	2.32489					
	Permanently sick	4.205729	.3354661	18.01	0.000	3.597046
>	4.917412					
	Homemaker	1.216866	.0665225	3.59	0.000	1.093227
>	1.354489					
	Other	1.857851	.1870434	6.15	0.000	1.525156
>	2.26312					
	bmi					
		1.050019	.0036296	14.12	0.000	1.042929
>	1.057157					
	changed					
	No longer officially together	.8646423	.1485342	-0.85	0.397	.6174634
>	1.21077					
	All other situations	1	(base)			
	Newly officially together	1.879802	.7292596	1.63	0.104	.8788055
>	4.020975					
	eurod					
		1.123952	.0075015	17.51	0.000	1.109345
>	1.138751					
	changed#c.eurod					
	No longer officially together	.9785441	.0317963	-0.67	0.504	.9181675
>	1.042891					
	Newly officially together	.9742585	.0971364	-0.26	0.794	.8013217
>	1.184517					
	country					
	Austria	1	(base)			
	Germany	.7137988	.0620967	-3.88	0.000	.6019021
>	.8464977					
	Sweden	.4041318	.0386928	-9.46	0.000	.3349858
>	.4875505					
	Spain	1.814524	.1413325	7.65	0.000	1.557625
>	2.113793					
	Italy	5.287941	.402838	21.86	0.000	4.554509
>	6.139479					
	France	1.266921	.1079111	2.78	0.005	1.072131
>	1.497102					
	Denmark	.6744273	.0652535	-4.07	0.000	.5579278

>	.8152528						
		Greece		.9500533	.0785659	-0.62	0.536
>	1.117221						.8078984
		Switzerland		.5090769	.0542528	-6.34	0.000
>	.6273311						.4131141
		Belgium		1.760046	.13883	7.17	0.000
>	2.054309						1.507934
		Czechia		1.183579	.0960379	2.08	0.038
>	1.387603						1.009553
		Poland		1.617032	.1529828	5.08	0.000
>	1.946473						1.343349
		_cons		6.82e-06	1.91e-06	-42.55	0.000
>	.0000118						3.95e-06
		/lnsig2u		.9186776	.0393149		.8416218
>	.9957335						
		sigma_u		1.583027	.0311183		1.523196
>	1.645208						
		rho		.4323745	.0096489		.4135702
>	.4513757						

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

175 * /// Mundlak_Sph_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Gi_* Mundlak_Ma
> r_* ///
>
176 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout

177 asdoc, text(xtlogitLim) save(report.doc) append
Click to Open File: report.doc

178 *asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Goo
> d Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_C_changed No Mundlak_C_chang
> ed All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Employed Mund
> lak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemaker Mundlak
> _Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_Oth_thinc_Above
> Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_eurod Mundlak_Gi_1
> Mundlak_Gi_2 Mundlak_Gi_3 Mundlak_Mar_mst_Officially Mundlak_Mar_mst_Away Mundlak_Ma
> r_mst_Never Mundlak_Mar_mst_Lost, save(report.doc) append

179
180 logit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed##c.eur
> od i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84194755
Iteration 2: Log pseudolikelihood = -78010964
Iteration 3: Log pseudolikelihood = -77790841
Iteration 4: Log pseudolikelihood = -77789818
Iteration 5: Log pseudolikelihood = -77789818

```

Logistic regression

```

Number of obs = 110,007
Wald chi2(37) = 5584.34
Prob > chi2   = 0.0000
Pseudo R2    = 0.2304

```

Log pseudolikelihood = -77789818

(Std. err. adjusted for **48,310** clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.059455	.0034042	17.97	0.000	1.052804
>	1.066148	yedu	.9836221	.0055529	-2.93	0.003	.9727986
>	.9945661	gender					
		Male	1 (base)				
>	1.151581	Female	1.051987	.0485508	1.10	0.272	.9610058
		thinc					
		No income	1 (base)				
		Below country average income	.4782865	.073734	-4.78	0.000	.353561
>	.6470112	Above country average income	.4014244	.0627918	-5.84	0.000	.2954315
>	.5454447	mst					
		Officially with partner	1 (base)				
>	1.44348	Away from partner	1.212664	.1078036	2.17	0.030	1.018757
>	1.463388	Never married	1.228286	.1097548	2.30	0.021	1.030954
>	1.253242	Lost partner	1.123588	.0626049	2.09	0.036	1.007347
		sphus					
		Excellent	1 (base)				
>	1.790155	Very good	1.351803	.1937132	2.10	0.035	1.02079
>	1.908172	Good	1.47709	.1929837	2.99	0.003	1.143396
>	4.126477	Fair	3.193801	.4175029	8.88	0.000	2.471931
>	15.6692	Poor	12.00211	1.632651	18.27	0.000	9.193237
		cjs					
		Retired	1 (base)				
>	1.312984	Employed or self-employed	1.113648	.0935604	1.28	0.200	.9445745
>	2.505773	Unemployed	1.796336	.3050601	3.45	0.001	1.287756
>	3.807604	Permanently sick	3.063933	.3396972	10.10	0.000	2.465511
>	1.352448	Homemaker	1.180292	.0819923	2.39	0.017	1.03005
>	1.831815	Other	1.452461	.1719625	3.15	0.002	1.151669
>	1.046403	bmi	1.037251	.004649	8.16	0.000	1.028179
		changed					
		No longer officially together	1.097341	.2631759	0.39	0.699	.6857998
>	1.755843	All other situations	1 (base)				
>	3.770363	Newly officially together	1.645456	.6961027	1.18	0.239	.7181076
>	1.096131	eurod	1.07834	.009003	9.03	0.000	1.060838

changed#c.eurod						
No longer officially together		.9704425	.0390438	-0.75	0.456	.8968574
> 1.050065						
Newly officially together		.9680485	.0925474	-0.34	0.734	.8026395
> 1.167545						
country						
Austria	1 (base)					
Germany		.8509982	.0739723	-1.86	0.063	.717693
> 1.009064						
Sweden		.4867902	.0505345	-6.93	0.000	.3971707
> .5966319						
Spain		1.755929	.165207	5.98	0.000	1.46023
> 2.111506						
Italy		3.912001	.2996614	17.81	0.000	3.366639
> 4.545708						
France		1.212807	.0991285	2.36	0.018	1.033282
> 1.423523						
Denmark		.7756956	.0725598	-2.72	0.007	.6457562
> .9317816						
Greece		1.027192	.0816644	0.34	0.736	.8789801
> 1.200396						
Switzerland		.6256717	.0667967	-4.39	0.000	.5075423
> .7712955						
Belgium		1.495425	.1196363	5.03	0.000	1.278402
> 1.749291						
Czechia		1.12234	.1194206	1.08	0.278	.9110741
> 1.382595						
Poland		1.423411	.1295455	3.88	0.000	1.190863
> 1.701369						
Qyear						
2015	1 (base)					
2017		.8009518	.0377903	-4.70	0.000	.7302058
> .8785521						
2020		1.028733	.0503088	0.58	0.562	.9347079
> 1.132217						
2022		1.058167	.0479857	1.25	0.212	.9681758
> 1.156524						
_cons		.0003829	.0001392	-21.64	0.000	.0001877
> .0007809						

Note: **_cons** estimates baseline odds.

```
181 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

182

183

```
184 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged
> c.eurod i.country
```

Fitting comparison model:

```
Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33347.304
Iteration 2: Log likelihood = -30411.484
Iteration 3: Log likelihood = -30168.81
Iteration 4: Log likelihood = -30167.543
Iteration 5: Log likelihood = -30167.543
```

Fitting full model:


```

tau = 0.0    Log likelihood = -30167.543
tau = 0.1    Log likelihood = -30008.398
tau = 0.2    Log likelihood = -29859.539
tau = 0.3    Log likelihood = -29733.391
tau = 0.4    Log likelihood = -29651.002
tau = 0.5    Log likelihood = -29649.359
tau = 0.6    Log likelihood = -29797.475

```

```

Iteration 0: Log likelihood = -29649.359
Iteration 1: Log likelihood = -29122.744
Iteration 2: Log likelihood = -29119.566
Iteration 3: Log likelihood = -29119.559
Iteration 4: Log likelihood = -29119.559

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-29119.559**

Wald chi2(30) = **7810.69**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.]
> interval]						
	age	.0917619	.0022403	40.96	0.000	.087371
> .0961528	yedu	-.0181178	.0040441	-4.48	0.000	-.0260441
> -.0101914	gender Female	.1165455	.0352111	3.31	0.001	.0475331
> .1855579	thinc					
Below country average income		-.5704357	.0950934	-6.00	0.000	-.7568153
> -.3840561	Above country average income	-.7157181	.0981184	-7.29	0.000	-.9080267
> -.5234095	single Yes	.1375728	.0373086	3.69	0.000	.0644492
> .2106964	sphus Very good	-.0367122	.0983679	-0.37	0.709	-.2295098
> .1560854	Good	.3506882	.0910401	3.85	0.000	.1722529
> .5291235	Fair	1.328479	.0917221	14.48	0.000	1.148707
> 1.508251	Poor	3.05736	.097629	31.32	0.000	2.86601
> 3.248709	cjs					
Employed or self-employed		.2987743	.0601642	4.97	0.000	.1808546
> .416694	Unemployed	.6113516	.1175621	5.20	0.000	.3809341
> .841769	Permanently sick	1.444181	.0765501	18.87	0.000	1.294146
> 1.594217	Homemaker	.1935064	.053061	3.65	0.000	.0895087
> .2975041	Other	.6138352	.0942831	6.51	0.000	.4290439

> .7986266						
	bmi	.0486491	.0031665	15.36	0.000	.0424428
> .0548553						
	changed					
	Hooked up	-.1197227	.2327475	-0.51	0.607	-.5758995
> .336454						
	Back single	-.2195829	.0784436	-2.80	0.005	-.3733295
> -.0658363						
	eurod	.1157354	.0063039	18.36	0.000	.1033801
> .1280908						
	country					
	Germany	-.3368096	.0884985	-3.81	0.000	-.5102633
> -.1633558						
	Sweden	-.9046483	.0981331	-9.22	0.000	-1.096986
> -.7123111						
	Spain	.5964797	.0801508	7.44	0.000	.4393869
> .7535724						
	Italy	1.663995	.0777802	21.39	0.000	1.511549
> 1.816442						
	France	.2395919	.0861992	2.78	0.005	.0706445
> .4085393						
	Denmark	-.389312	.098851	-3.94	0.000	-.5830563
> -.1955677						
	Greece	-.0578875	.0828922	-0.70	0.485	-.2203533
> .1045782						
	Switzerland	-.6735124	.1069446	-6.30	0.000	-.88312
> -.4639048						
	Belgium	.564652	.081017	6.97	0.000	.4058615
> .7234424						
	Czechia	.1598944	.0834207	1.92	0.055	-.0036071
> .323396						
	Poland	.4737736	.0958686	4.94	0.000	.2858745
> .6616726						
	_cons	-11.82667	.2603326	-45.43	0.000	-12.33691
> -11.31643						
	/lnsig2u	.9199512	.0386758			.8441481
> .9957544						
	sigma_u	1.584035	.0306319			1.525121
> 1.645225						
	rho	.4326871	.0094937			.414183
> .4513808						

LR test of rho=0: chibar2(01) = 2095.97

Prob >= chibar2 = 0.000

185 estimates store RE

186 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ibl.schanged
> c.eurod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or all negative outcomes.

note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5684.9537**
 Iteration 1: Log likelihood = **-5284.3129**
 Iteration 2: Log likelihood = **-5282.3016**
 Iteration 3: Log likelihood = **-5282.301**
 Iteration 4: Log likelihood = **-5282.301**

Conditional fixed-effects logistic regression
 Group variable: **id**

Number of obs = **17,262**
 Number of groups = **5,962**

Obs per group:
 min = **2**
 avg = **2.9**
 max = **4**

Log likelihood = **-5282.301**

LR chi2(18) = **2033.13**
 Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.]
> interval]						
	age	.1736827	.0061113	28.42	0.000	.1617047
> .1856606	yedu	-.041587	.0195352	-2.13	0.033	-.0798752
> -.0032989	gender Female	0 (omitted)				
	thinc					
Below country average income		.1518173	.1266738	1.20	0.231	-.0964587
> .4000933	Above country average income	.1023607	.1344671	0.76	0.447	-.16119
> .3659115	single Yes	.0237966	.142113	0.17	0.867	-.2547397
> .3023329	sphus Very good	-.2041543	.1434483	-1.42	0.155	-.4853079
> .0769992	Good	-.0334229	.1388319	-0.24	0.810	-.3055284
> .2386826	Fair	.4375895	.1418472	3.08	0.002	.1595741
> .715605	Poor	1.375489	.150051	9.17	0.000	1.081394
> 1.669583	cjs Employed or self-employed	.5093894	.1030404	4.94	0.000	.3074339
> .7113448	Unemployed	.5618251	.1761819	3.19	0.001	.216515
> .9071353	Permanently sick	.761857	.1259498	6.05	0.000	.5149998
> 1.008714	Homemaker	.1334421	.0985912	1.35	0.176	-.0597932
> .3266773	Other	.4734759	.1416124	3.34	0.001	.1959207
> .7510312	bmi	-.0029733	.0075599	-0.39	0.694	-.0177905
> .0118438	schanged Hooked up	-.2134335	.2902511	-0.74	0.462	-.7823153
> .3554483	Back single	-.1278368	.1386576	-0.92	0.357	-.3996007

> .1439271

> .0947463

eurod	.0762879	.0094177	8.10	0.000	.0578295
country					
Germany	0	(omitted)			
Sweden	0	(omitted)			
Spain	0	(omitted)			
Italy	0	(omitted)			
France	0	(omitted)			
Denmark	0	(omitted)			
Greece	0	(omitted)			
Switzerland	0	(omitted)			
Belgium	0	(omitted)			
Czechia	0	(omitted)			
Poland	0	(omitted)			

187 estimates store FE

188 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
age	.1736827	.0917619	.0819207	.0056859
yedu	-.041587	-.0181178	-.0234693	.019112
thinc				
1	.1518173	-.5704357	.722253	.0836869
2	.1023607	-.7157181	.8180788	.0919466
1.single	.0237966	.1375728	-.1137762	.1371283
sphus				
2	-.2041543	-.0367122	-.1674421	.1044087
3	-.0334229	.3506882	-.3841111	.1048141
4	.4375895	1.328479	-.8908899	.1082021
5	1.375489	3.05736	-1.681871	.1139469
cjs				
2	.5093894	.2987743	.2106151	.0836516
3	.5618251	.6113516	-.0495264	.131222
4	.761857	1.444181	-.6823245	.1000172
5	.1334421	.1935064	-.0600643	.0830949
6	.4734759	.6138352	-.1403593	.1056635
bmi	-.0029733	.0486491	-.0516224	.0068648
schanged				
0	-.2134335	-.1197227	-.0937108	.1734195
2	-.1278368	-.2195829	.0917461	.1143352
eurod	.0762879	.1157354	-.0394475	.0069968

b = Consistent under H0 and Ha; obtained from **xtlogit**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(18) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = **1179.39**
 Prob > chi2 = **0.0000**
 Click to Open File: [report.doc](#)

189

190

191 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ibl.schanged#

> #c.eurod i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.Iteration 0: Log likelihood = **-5682.7646**Iteration 1: Log likelihood = **-5282.4307**Iteration 2: Log likelihood = **-5280.405**Iteration 3: Log likelihood = **-5280.4044**Iteration 4: Log likelihood = **-5280.4044**

Conditional fixed-effects logistic regression

Group variable: **id**Number of obs = **17,262**Number of groups = **5,962**

Obs per group:

min = **2**avg = **2.9**max = **4**LR chi2(20) = **2036.93**Prob > chi2 = **0.0000**Log likelihood = **-5280.4044**

> interval]		phinact	Odds ratio	Std. err.	z	P> z	[95% conf.]
		age	1.189626	.0072737	28.40	0.000	1.175455
> 1.203968		yedu	.9592752	.0187274	-2.13	0.033	.9232636
> .9966915		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		thinc					
		No income	1	(base)			
Below country average income			1.159993	.1469164	1.17	0.241	.9049996
> 1.486834		Above country average income	1.105394	.1486059	0.75	0.456	.8493446
> 1.438633		single					
		No	1	(base)			
> 1.351981		Yes	1.022438	.1457425	0.16	0.876	.7732201
		sphus					
		Excellent	1	(base)			
> 1.086753		Very good	.8201882	.1177644	-1.38	0.167	.6190079
> 1.275963		Good	.971781	.1350236	-0.21	0.837	.740114
> 2.054956		Fair	1.555829	.220873	3.11	0.002	1.177934

	Poor	3.966017	.5955585	9.17	0.000	2.954837
> 5.323234						
	cjs					
	Retired	1	(base)			
	Employed or self-employed	1.671461	.1723108	4.98	0.000	1.365671
> 2.045722						
	Unemployed	1.745641	.3079349	3.16	0.002	1.235382
> 2.466657						
	Permanently sick	2.138735	.2694218	6.03	0.000	1.670818
> 2.737695						
	Homemaker	1.139352	.112433	1.32	0.186	.9389882
> 1.38247						
	Other	1.606685	.2275684	3.35	0.001	1.217217
> 2.12077						
	bmi	.9972502	.0075442	-0.36	0.716	.9825729
> 1.012147						
	schanged					
	Hooked up	.4670907	.22128	-1.61	0.108	.1845662
> 1.182089						
	Unchanged	1	(base)			
	Back single	1.052301	.2159861	0.25	0.804	.7037689
> 1.57344						
	eurod	1.080849	.0104042	8.08	0.000	1.060648
> 1.101434						
	schanged#c.eurod					
	Hooked up	1.182463	.135194	1.47	0.143	.9450776
> 1.479475						
	Back single	.9616233	.0321645	-1.17	0.242	.9006039
> 1.026777						
	country					
	Austria	1	(base)			
	Germany	1	(omitted)			
	Sweden	1	(omitted)			
	Spain	1	(omitted)			
	Italy	1	(omitted)			
	France	1	(omitted)			
	Denmark	1	(omitted)			
	Greece	1	(omitted)			
	Switzerland	1	(omitted)			
	Belgium	1	(omitted)			
	Czechia	1	(omitted)			
	Poland	1	(omitted)			

```

192 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout

```

193

```

194 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged#
> #c.eurod i.country, or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33347.355
Iteration 2: Log pseudolikelihood = -30410.942
Iteration 3: Log pseudolikelihood = -30168.151
Iteration 4: Log pseudolikelihood = -30166.882
Iteration 5: Log pseudolikelihood = -30166.882

```

Fitting full model:

```

tau = 0.0    Log pseudolikelihood = -30166.882
tau = 0.1    Log pseudolikelihood = -30007.7
tau = 0.2    Log pseudolikelihood = -29858.796
tau = 0.3    Log pseudolikelihood = -29732.592
tau = 0.4    Log pseudolikelihood = -29650.132
tau = 0.5    Log pseudolikelihood = -29648.399
tau = 0.6    Log pseudolikelihood = -29796.396

```

```

Iteration 0: Log pseudolikelihood = -29648.399
Iteration 1: Log pseudolikelihood = -29121.358
Iteration 2: Log pseudolikelihood = -29118.097
Iteration 3: Log pseudolikelihood = -29118.09
Iteration 4: Log pseudolikelihood = -29118.09

```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Number of obs = 110,045
Number of groups = 48,341

Random effects u_i ~ Gaussian

Obs per group:
min = 1
avg = 2.3
max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = **-29118.09**

Wald chi2(32) = 7574.56
Prob > chi2 = 0.0000

(Std. err. adjusted for 48,341 clusters)

> rs in **id**)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.]
> interval]						
	age	1.09612	.0025959	38.75	0.000	1.091044
> 1.101219	yedu	.9820448	.0040239	-4.42	0.000	.9741897
> .9899632	gender					
	Male	1 (base)				
> 1.204309	Female	1.123175	.0399688	3.26	0.001	1.047507
	thinc					
	No income	1 (base)				
	Below country average income	.5648303	.0583067	-5.53	0.000	.4613701
> .6914909	Above country average income	.4889057	.0519638	-6.73	0.000	.396967
> .6021377	single					
	No	1 (base)				
> 1.236277	Yes	1.147484	.0436362	3.62	0.000	1.065068
	sphus					
	Excellent	1 (base)				
> 1.168028	Very good	.9641822	.0943496	-0.37	0.709	.7959122
> 1.700078	Good	1.419875	.1304753	3.82	0.000	1.185854
> 4.522503	Fair	3.772586	.3489802	14.35	0.000	3.147019
> 25.80468	Poor	21.25095	2.105125	30.85	0.000	17.5008
	cjs					

	Retired	1 (base)				
> 1.521774	Employed or self-employed	1.348672	.0830936	4.85	0.000	1.195261
> 2.335156	Unemployed	1.842945	.222581	5.06	0.000	1.454484
> 4.950982	Permanently sick	4.234199	.3378603	18.09	0.000	3.621188
> 1.351639	Homemaker	1.214451	.0663164	3.56	0.000	1.091187
> 2.252185	Other	1.84937	.1859368	6.12	0.000	1.5186
> 1.056992	bmi	1.049859	.003627	14.08	0.000	1.042774
> 1.429573	schanged Hooked up	.5959066	.2660489	-1.16	0.246	.2483992
> 1.226878	Unchanged Back single	1 (base) .9238282	.1337226	-0.55	0.584	.6956346
> 1.138731	eurod	1.123864	.0075357	17.42	0.000	1.109191
> 1.351729	schanged#c.eurod Hooked up	1.119933	.1074899	1.18	0.238	.9278864
> 1.024105	Back single	.9683625	.0276523	-1.13	0.260	.9156538
> .8470172	country Austria	1 (base) .7142015	.0621497	-3.87	0.000	.6022119
> .4880919	Germany	.4044722	.0387807	-9.44	0.000	.3351781
> 2.114999	Sweden	1.815765	.1413242	7.66	0.000	1.558867
> 6.13234	Spain	5.283352	.4016925	21.89	0.000	4.551901
> 1.500983	Italy	1.270059	.1082529	2.80	0.005	1.074663
> .819168	France	.6775263	.0656247	-4.02	0.000	.5603758
> 1.110744	Denmark	.9447548	.0780209	-0.69	0.491	.8035708
> .6283443	Greece	.5098774	.0543493	-6.32	0.000	.4137459
> 2.053085	Switzerland	1.758781	.138841	7.15	0.000	1.506665
> 1.375312	Belgium	1.173429	.0950436	1.97	0.048	1.001181
> 1.932946	Czechia	1.606384	.1516746	5.02	0.000	1.334993
> .0000125	Poland	7.28e-06	2.00e-06	-42.95	0.000	4.24e-06
> .9978228	_cons	.920913	.0392404			.8440031
> 1.646927	/lnsig2u	1.584797	.0310941			1.525011
> .4518931	sigma_u	.4329232	.0096335			.4141479
	rho					

Note: Estimates are transformed only in the first equation to odds ratios.
Note: _cons estimates baseline odds (conditional on zero random effects).


```

195 * /// Mundlak_Sph_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_sGi_* Mundlak_
> Sin_* ///
>
196 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *)
> ctitle(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results2.xls
  dir : seeout

197 asdoc, text(xtlogitLim) save(report.doc) append
  Click to Open File: report.doc

198 *asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Goo
> d Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_sC_schanged_Hooked Mundlak_s
> C_schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired Mundlak_Oth_cj
> s_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Ho
> memaker Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_O
> th_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_euro
> d Mundlak_sGi_1 Mundlak_sGi_2 Mundlak_sGi_3 Mundlak_Sin_single_No Mundlak_Sin_single
> _Yes, save(report.doc) append

199
200 logit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged##c
> .eurod i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84158818
Iteration 2: Log pseudolikelihood = -77976735
Iteration 3: Log pseudolikelihood = -77756657
Iteration 4: Log pseudolikelihood = -77755617
Iteration 5: Log pseudolikelihood = -77755617

```

Logistic regression

Number of obs = 110,007

Wald chi2(35) = 5558.65

Prob > chi2 = 0.0000

Pseudo R2 = 0.2308

Log pseudolikelihood = -77755617

(Std. err. adjusted for 48,310 cluste

> rs in id)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.057819	.0033249	17.88	0.000	1.051322
> 1.064356	yedu	.983553	.0055296	-2.95	0.003	.9727747
> .9944507	gender					
	Male	1 (base)				
> 1.131545	Female	1.033548	.0477688	0.71	0.475	.9440386
	thinc					
	No income	1 (base)				
Below country average income		.4780525	.0731591	-4.82	0.000	.3541696
> .6452675	Above country average income	.4076607	.0631077	-5.80	0.000	.300974
> .5521648	single					
	No	1 (base)				
> 1.334766	Yes	1.207747	.0616203	3.70	0.000	1.092816
	sphus					
	Excellent	1 (base)				
> 1.781065	Very good	1.344607	.1928516	2.06	0.039	1.015105
	Good	1.473733	.1926859	2.97	0.003	1.140583

> 1.904192						
> 4.121022	Fair	3.189276	.4170661	8.87	0.000	2.468194
> 15.67506	Poor	12.00902	1.632356	18.29	0.000	9.200381
	cjs					
	Retired	1	(base)			
> 1.309314	Employed or self-employed	1.109913	.0935637	1.24	0.216	.9408802
> 2.505137	Unemployed	1.80146	.3030793	3.50	0.000	1.295441
> 3.804856	Permanently sick	3.064604	.3383006	10.15	0.000	2.468372
> 1.349723	Homemaker	1.178312	.0816517	2.37	0.018	1.028669
> 1.79757	Other	1.42721	.1680018	3.02	0.003	1.133157
> 1.046318	bmi	1.037215	.0046247	8.19	0.000	1.02819
> 1.414	schanged Hooked up	.5719507	.2641308	-1.21	0.226	.2313491
> 1.357241	Unchanged Back single	1 .9248256	(base) .1810067	-0.40	0.690	.6301774
> 1.096324	eurod	1.078442	.0090489	9.00	0.000	1.060851
> 1.173187	schanged#c.eurod Hooked up	.98425	.0881821	-0.18	0.859	.8257405
> 1.042323	Back single	.9729173	.0342058	-0.78	0.435	.9081329
> 1.008263	country Austria	1 .8503317	(base) .0739097	-1.87	0.062	.7171386
> .5991395	Germany	.4883652	.0509382	-6.87	0.000	.3980718
> 2.092146	Sweden	1.740039	.1636047	5.89	0.000	1.447191
> 4.495661	Spain	3.872952	.2946181	17.80	0.000	3.336496
> 1.420141	Italy	1.209747	.0989694	2.33	0.020	1.030523
> .936591	France	.779601	.0729753	-2.66	0.008	.6489255
> 1.186134	Denmark	1.015368	.0805303	0.19	0.848	.8691879
> .7730681	Greece	.6270144	.0669886	-4.37	0.000	.5085543
> 1.756224	Switzerland	1.500648	.1204133	5.06	0.000	1.282264
> 1.373712	Belgium	1.115594	.1184656	1.03	0.303	.9059752
> 1.667675	Czechia	1.397131	.1261785	3.70	0.000	1.170477
> .8905923	Poland	.8119667	.0382905	-4.42	0.000	.7402827
> 1.151374	Qyear 2015	1	(base)			
> 1.177215	2017	1.045917	.0512625	0.92	0.360	.9501196
	2020	1.076896	.0489386	1.63	0.103	.985126
	2022					

	_cons		.0004273	.0001523	-21.76	0.000	.0002124
> .0008593							

Note: **_cons** estimates baseline odds.

```
201 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

202

```
203 log close InactivityAndMarriage
      name: InactivityAndMarriage
      log: C:\Users\hansg\Documents\working\SHARE data 5\output.smcl
      log type: smcl
      closed on: 30 Jun 2026, 12:00:30
```
